

AviLight Ecological Impact and Diagnostics Report

Executive Summary

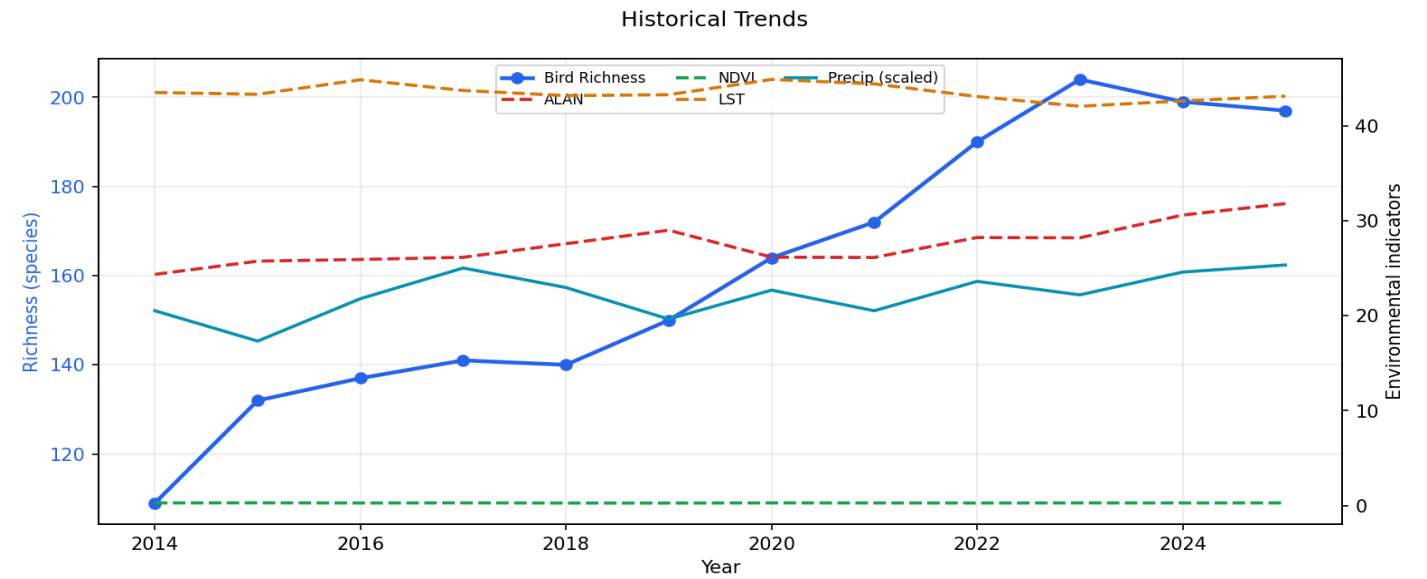
Run Context

Field	Value
Area	All Areas
Trend Years	2014-2025
Snapshot	2025-01
Generated	2026-04-19 16:16:52 UTC

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Ecological Impact View

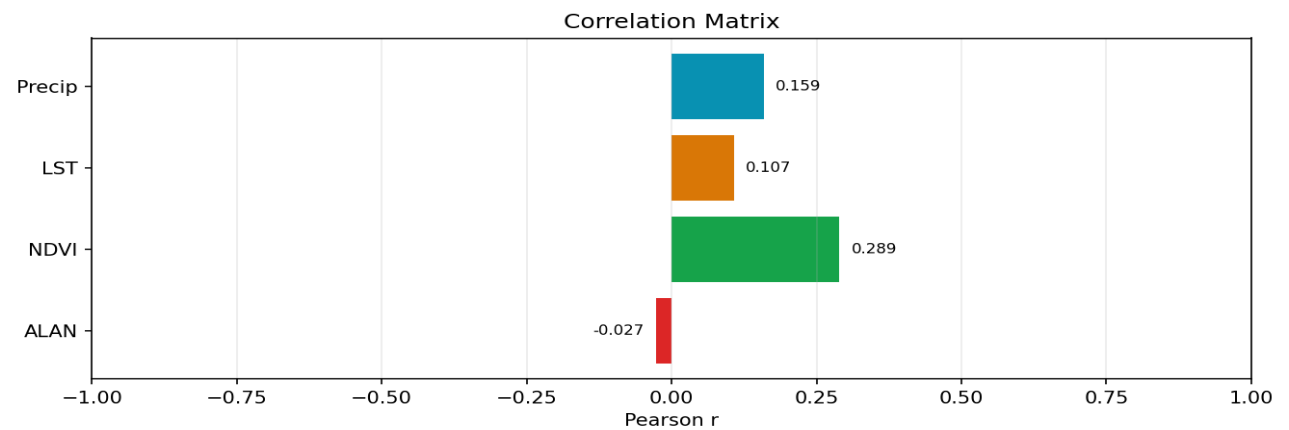
Chart: Long-Term Ecological Impact Trends



Numerical Values: Historical Trends

Year	Richness	ALAN	NDVI	LST	Precip
2014	109.00	24.35	0.2770	43.51	2052.59
2015	132.00	25.75	0.2874	43.32	1732.52
2016	137.00	25.92	0.2711	44.86	2179.36
2017	141.00	26.15	0.2829	43.73	2502.38
2018	140.00	27.58	0.2659	43.19	2296.71
2019	150.00	29.01	0.2643	43.28	1964.26
2020	164.00	26.17	0.2801	44.89	2269.63
2021	172.00	26.14	0.2759	44.42	2052.01
2022	190.00	28.24	0.2698	43.09	2361.96
2023	204.00	28.21	0.2786	42.06	2219.41
2024	199.00	30.60	0.2762	42.63	2459.51
2025	197.00	31.81	0.2805	43.12	2534.40

Chart: Correlation Matrix

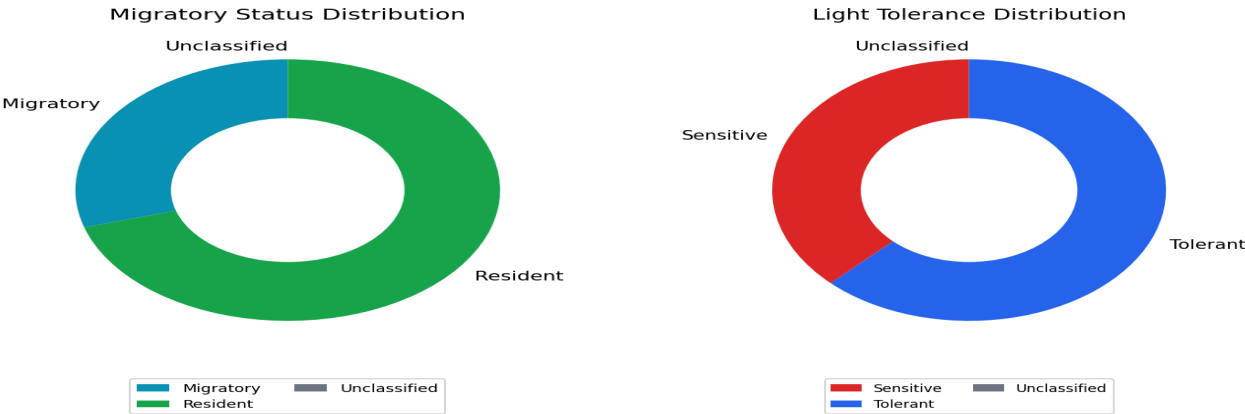


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Numerical Values: Correlation Coefficients

Metric	Pearson r
ALAN vs Richness	-0.0271
NDVI vs Richness	0.2895
LST vs Richness	0.1075
Precipitation vs Richness	0.1587

Chart: Migratory and Light Tolerance Distribution



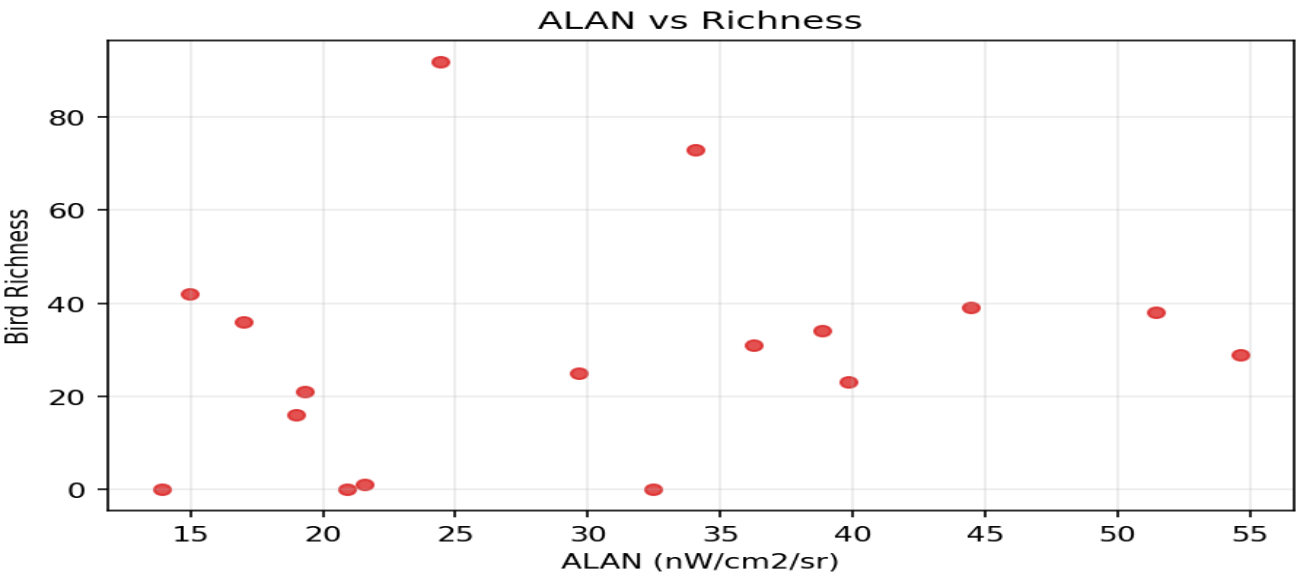
Numerical Values: Migration Distribution

Category	Count
Migratory	37
Resident	88
Unclassified	0

Numerical Values: Light Tolerance Distribution

Category	Count
Sensitive	47
Tolerant	78
Unclassified	0

Chart: ALAN vs Richness

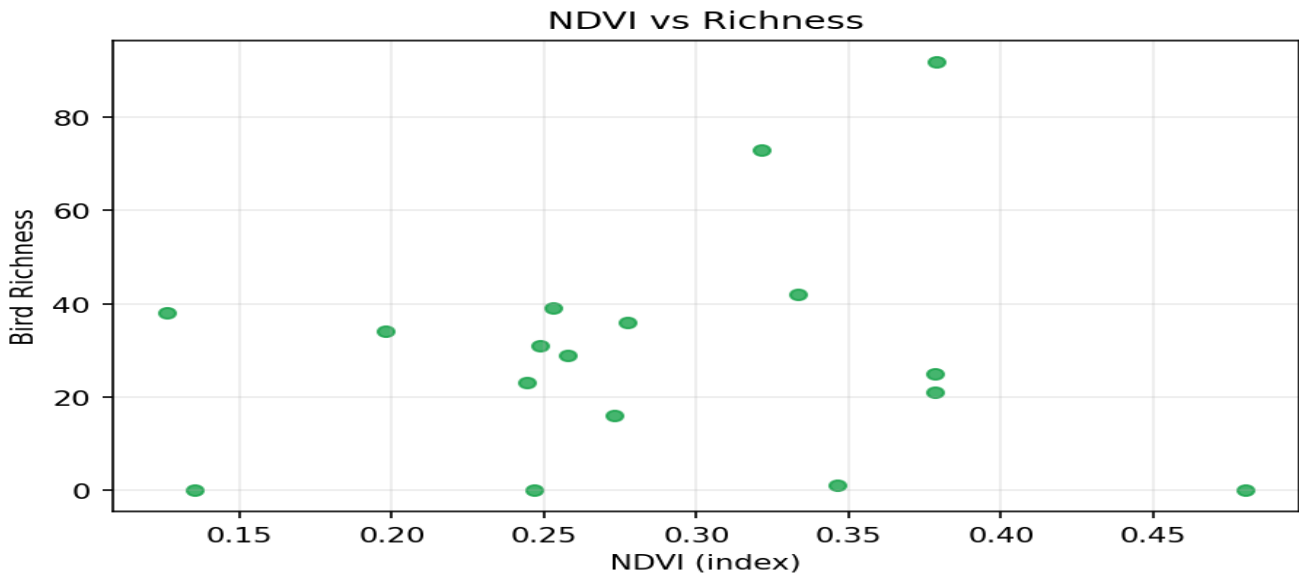


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Numerical Values: ALAN vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	14.9351	42.00
Las Piñas	Las Piñas	19.2981	21.00
Makati	Makati	44.4308	39.00
Malabon	Malabon	18.9817	16.00
Mandaluyong	Mandaluyong	39.8210	23.00
Manila	Manila	51.4294	38.00
Marikina	Marikina	21.5588	1.00
Muntinlupa	Muntinlupa	20.9031	0.00
Navotas	Navotas	13.8866	0.00
Parañaque	Parañaque	34.0353	73.00
Pasay	Pasay	54.6484	29.00
Pasig	Pasig	36.2831	31.00
Pateros	Pateros	32.4703	0.00
Quezon City	Quezon City	24.4084	92.00
San Juan	San Juan	38.8550	34.00
Taguig	Taguig	29.6295	25.00
Valenzuela	Valenzuela	16.9789	36.00

Chart: NDVI vs Richness



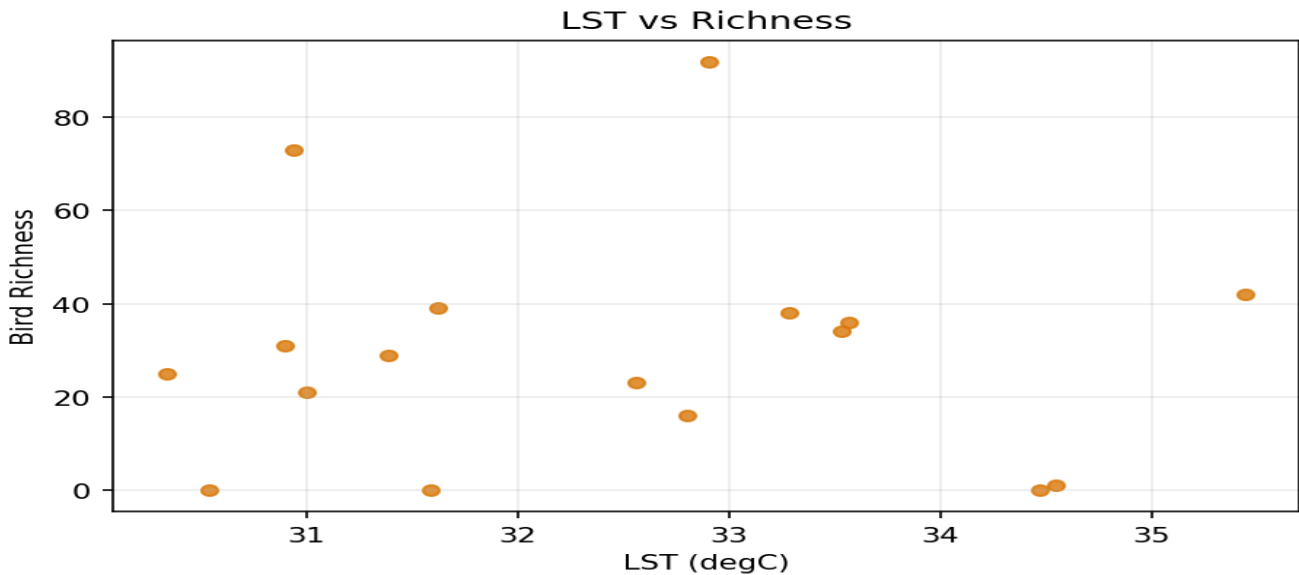
Numerical Values: NDVI vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	0.3331	42.00
Las Piñas	Las Piñas	0.3782	21.00
Makati	Makati	0.2530	39.00
Malabon	Malabon	0.2731	16.00
Mandaluyong	Mandaluyong	0.2444	23.00
Manila	Manila	0.1259	38.00
Marikina	Marikina	0.3463	1.00
Muntinlupa	Muntinlupa	0.4803	0.00
Navotas	Navotas	0.1350	0.00

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Parañaque	Parañaque	0.3212	73.00
Pasay	Pasay	0.2579	29.00
Pasig	Pasig	0.2488	31.00
Pateros	Pateros	0.2468	0.00
Quezon City	Quezon City	0.3790	92.00
San Juan	San Juan	0.1977	34.00
Taguig	Taguig	0.3785	25.00
Valenzuela	Valenzuela	0.2773	36.00

Chart: LST vs Richness

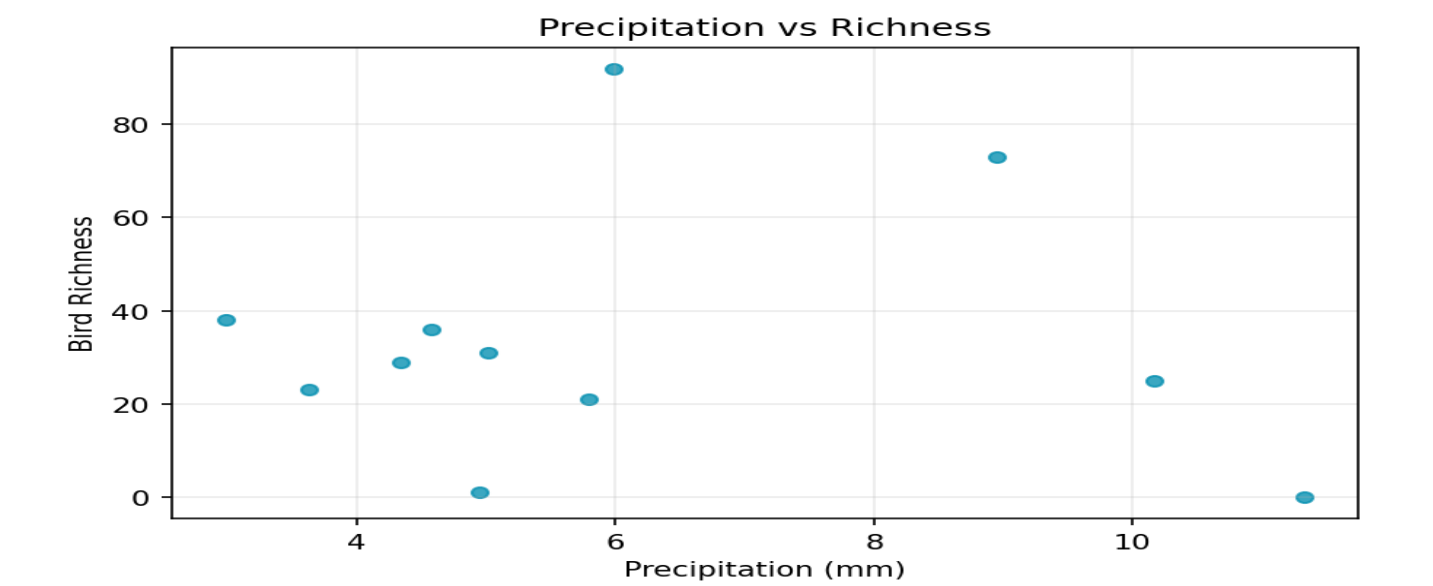


Numerical Values: LST vs Richness

Site	Area	X	Richness
Caloocan	Caloocan	35.4455	42.00
Las Piñas	Las Piñas	30.9998	21.00
Makati	Makati	31.6204	39.00
Malabon	Malabon	32.8041	16.00
Mandaluyong	Mandaluyong	32.5610	23.00
Manila	Manila	33.2872	38.00
Marikina	Marikina	34.5482	1.00
Muntinlupa	Muntinlupa	30.5393	0.00
Navotas	Navotas	31.5857	0.00
Parañaque	Parañaque	30.9411	73.00
Pasay	Pasay	31.3898	29.00
Pasig	Pasig	30.9011	31.00
Pateros	Pateros	34.4750	0.00
Quezon City	Quezon City	32.9052	92.00
San Juan	San Juan	33.5332	34.00
Taguig	Taguig	30.3364	25.00
Valenzuela	Valenzuela	33.5701	36.00

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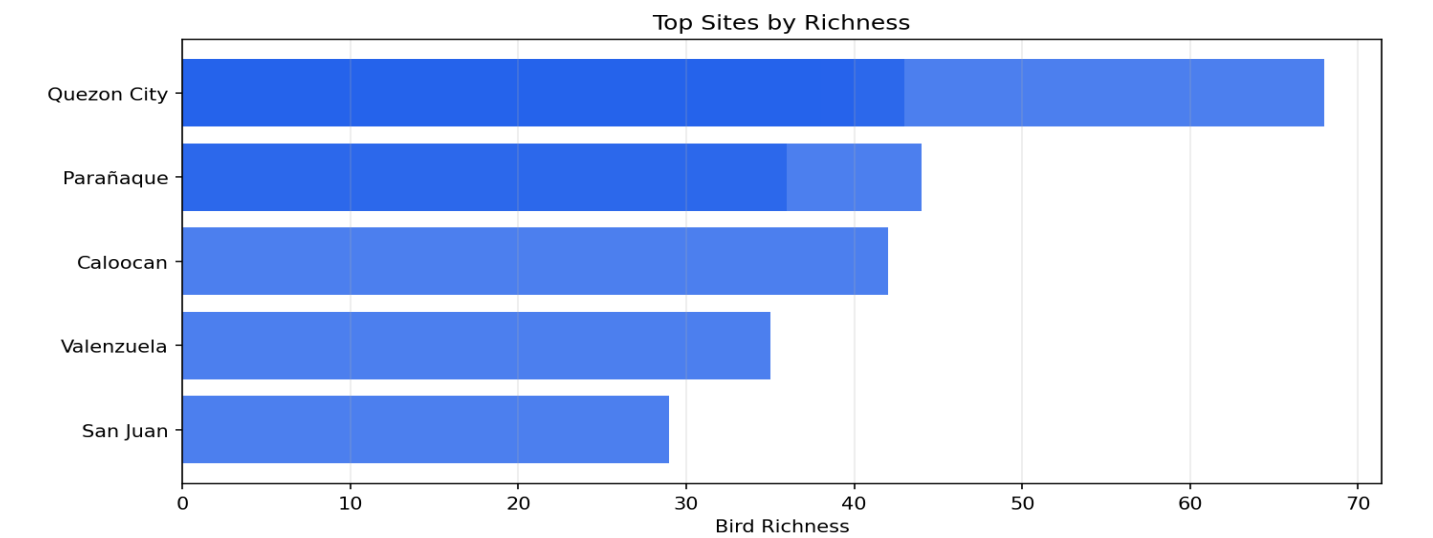
Chart: Precipitation vs Richness



Numerical Values: Precipitation vs Richness

Site	Area	X	Richness
Las Piñas	Las Piñas	5.7889	21.00
Mandaluyong	Mandaluyong	3.6254	23.00
Manila	Manila	2.9811	38.00
Marikina	Marikina	4.9446	1.00
Muntinlupa	Muntinlupa	11.3271	0.00
Parañaque	Parañaque	8.9451	73.00
Pasay	Pasay	4.3336	29.00
Pasig	Pasig	5.0141	31.00
Quezon City	Quezon City	5.9799	92.00
Taguig	Taguig	10.1621	25.00
Valenzuela	Valenzuela	4.5755	36.00

Chart: Top 10 Sites by Richness



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Numerical Values: Top Sites by Richness

Rank	Area	Richness
1	Quezon City	68
2	Parañaque	44
3	Quezon City	43
4	Caloocan	42
5	Quezon City	40
6	Quezon City	38
7	Parañaque	36
8	Valenzuela	35
9	Quezon City	34
10	San Juan	29

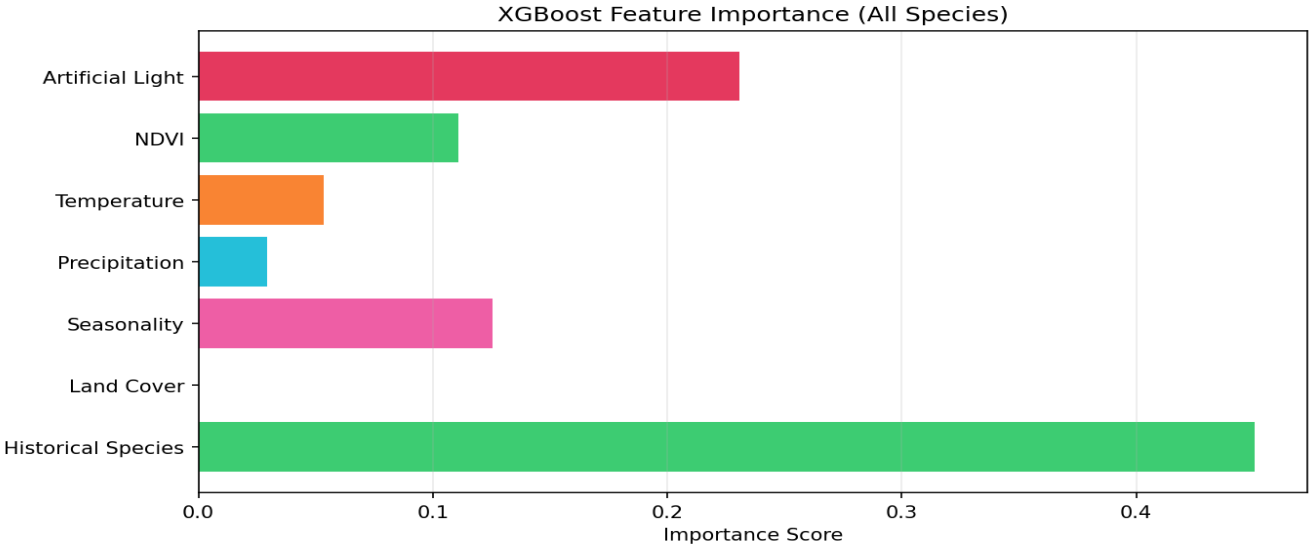
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Model Diagnostics View

KPI Cards

Metric	Value
RMSE	6.3924
MAE	4.0568
R2	0.7445

Chart: XGBoost Feature Importance (All Species)

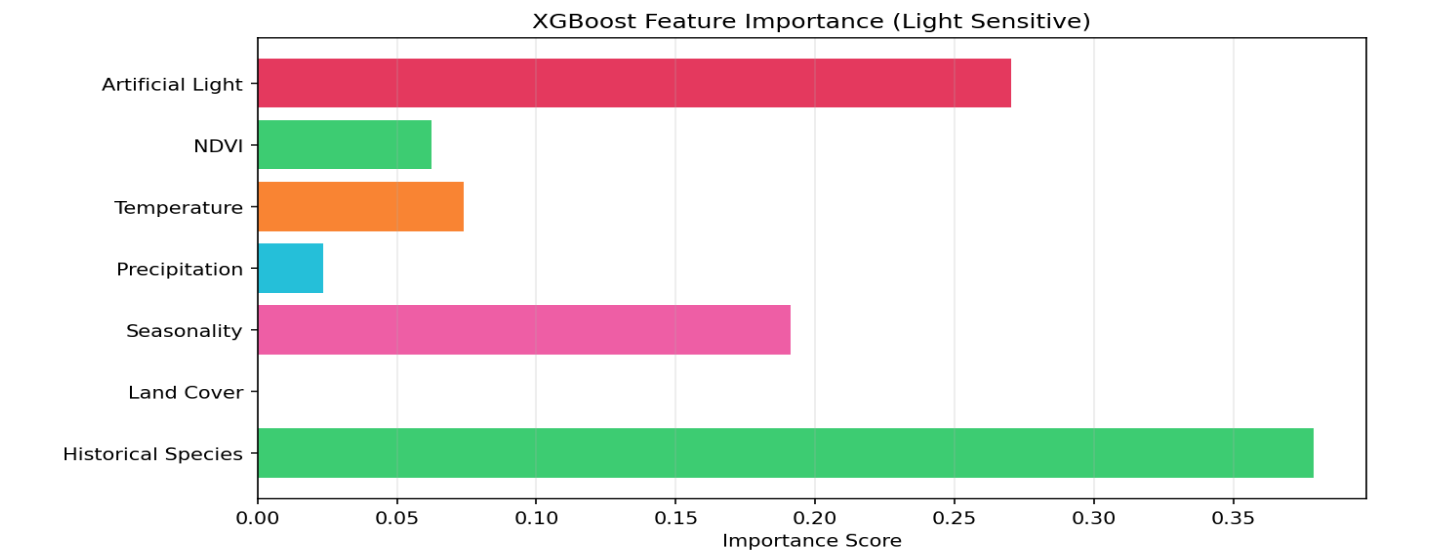


Numerical Values: XGBoost Feature Importance (All Species)

Feature	Importance
Artificial Light	0.230796
NDVI	0.110724
Temperature	0.053409
Precipitation	0.029266
Seasonality	0.125400
Land Cover	0.000000
Historical Species	0.450406

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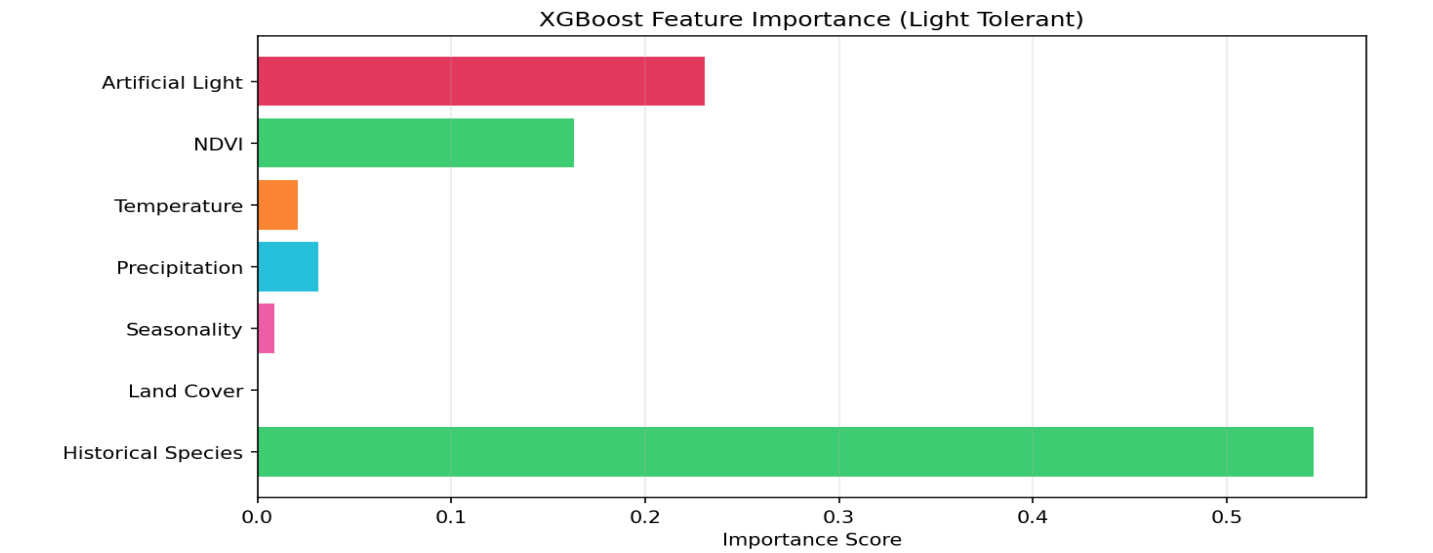
Chart: XGBoost Feature Importance (Light Sensitive)



Numerical Values: XGBoost Feature Importance (Light Sensitive)

Feature	Importance
Artificial Light	0.270363
NDVI	0.062340
Temperature	0.073999
Precipitation	0.023678
Seasonality	0.191019
Land Cover	0.000000
Historical Species	0.378602

Chart: XGBoost Feature Importance (Light Tolerant)



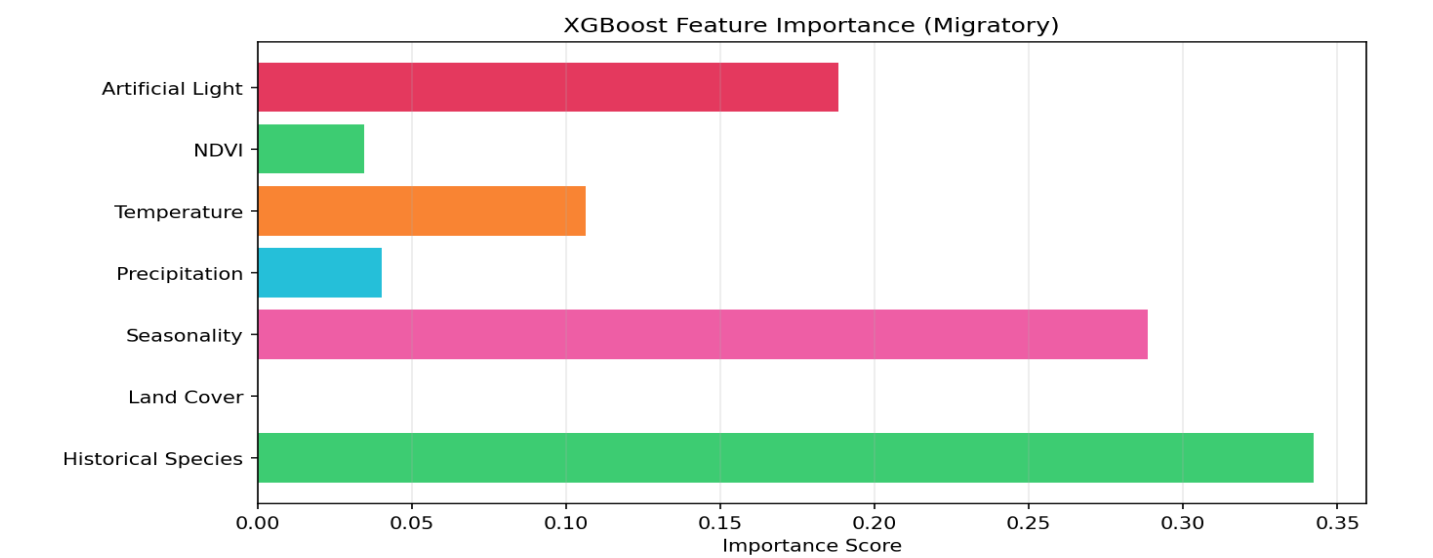
Numerical Values: XGBoost Feature Importance (Light Tolerant)

Feature	Importance
Artificial Light	0.230950
NDVI	0.163174
Temperature	0.020697

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Precipitation	0.031523
Seasonality	0.008989
Land Cover	0.000000
Historical Species	0.544666

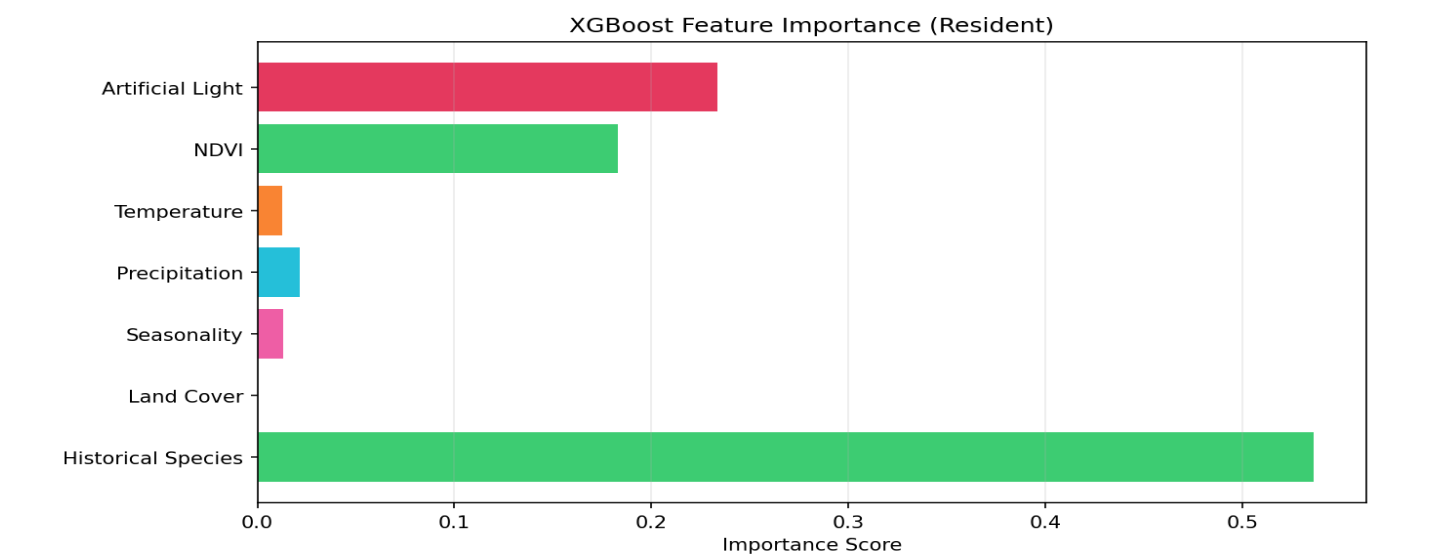
Chart: XGBoost Feature Importance (Migratory)



Numerical Values: XGBoost Feature Importance (Migratory)

Feature	Importance
Artificial Light	0.188284
NDVI	0.034494
Temperature	0.106226
Precipitation	0.040161
Seasonality	0.288632
Land Cover	0.000000
Historical Species	0.342202

Chart: XGBoost Feature Importance (Resident)

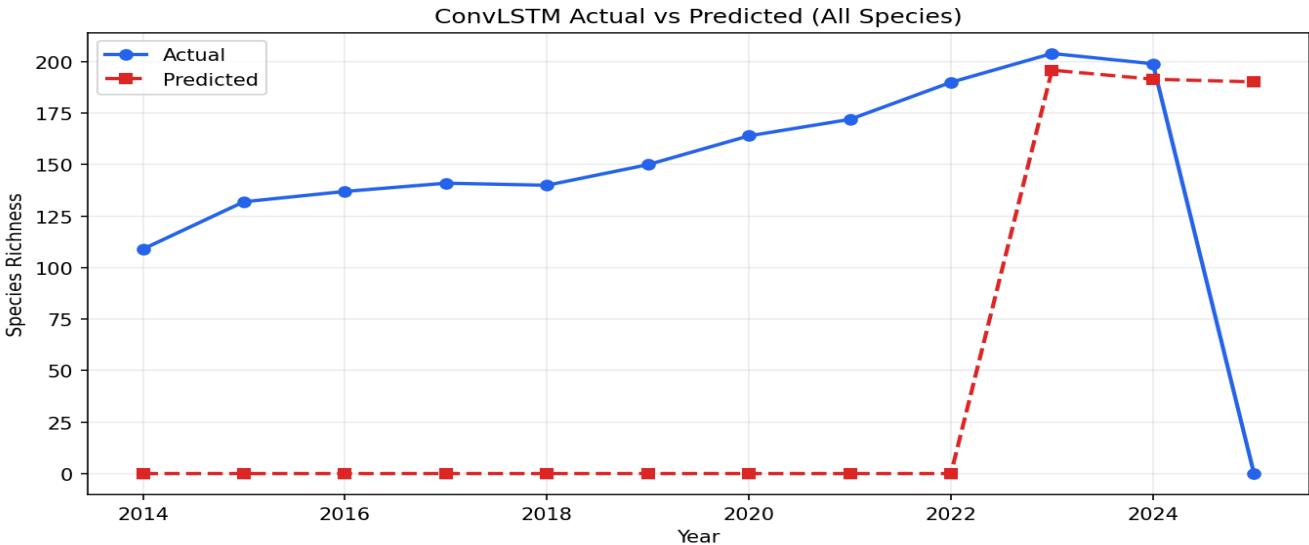


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Numerical Values: XGBoost Feature Importance (Resident)

Feature	Importance
Artificial Light	0.233587
NDVI	0.182887
Temperature	0.012712
Precipitation	0.021700
Seasonality	0.012960
Land Cover	0.000000
Historical Species	0.536153

Chart: ConvLSTM Actual vs Predicted (All Species)

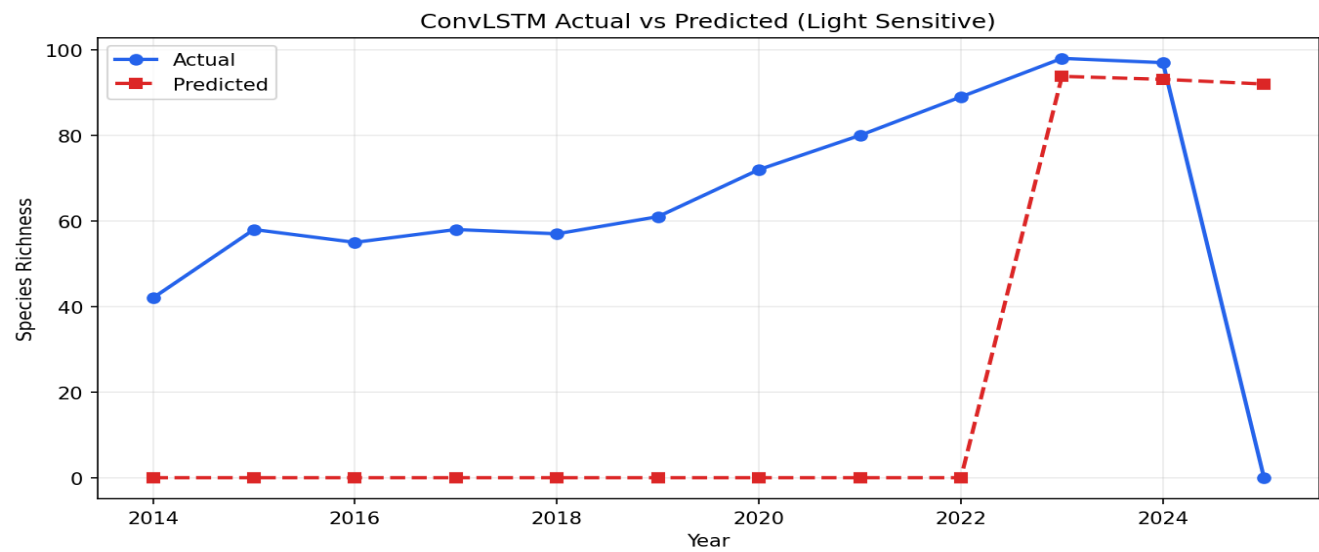


Numerical Values: ConvLSTM Actual vs Predicted (All Species)

Year	Actual	Predicted
2014	109.0000	0.0000
2015	132.0000	0.0000
2016	137.0000	0.0000
2017	141.0000	0.0000
2018	140.0000	0.0000
2019	150.0000	0.0000
2020	164.0000	0.0000
2021	172.0000	0.0000
2022	190.0000	0.0000
2023	204.0000	195.9000
2024	199.0000	191.5000
2025	0.0000	190.2000

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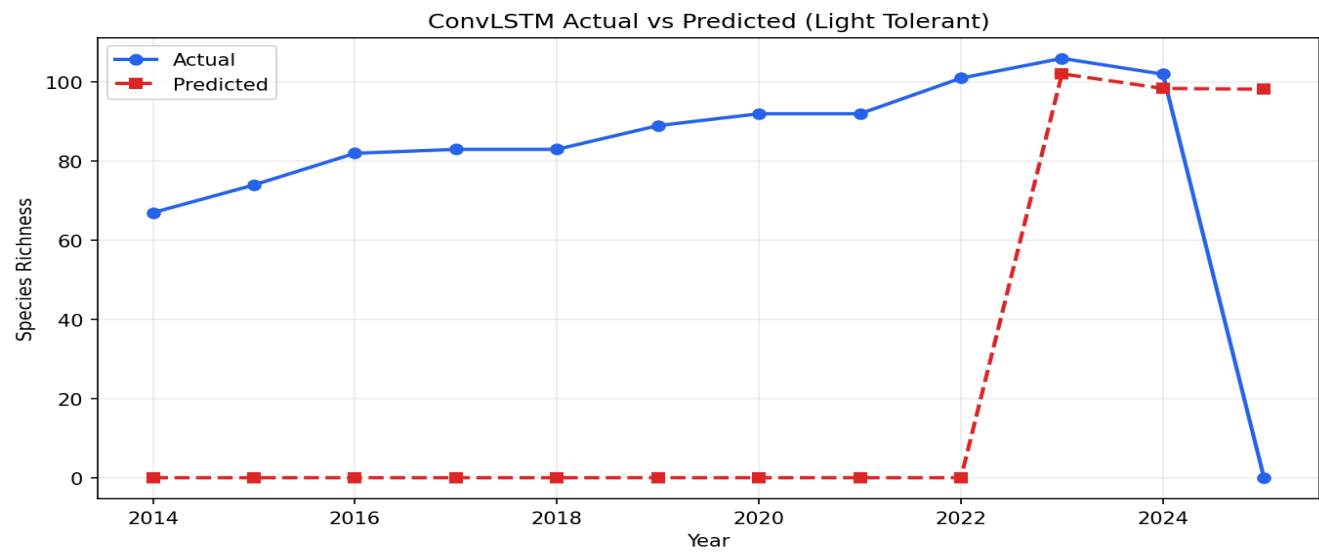
Chart: ConvLSTM Actual vs Predicted (Light Sensitive)



Numerical Values: ConvLSTM Actual vs Predicted (Light Sensitive)

Year		Actual	Predicted
	2014	42.0000	0.0000
	2015	58.0000	0.0000
	2016	55.0000	0.0000
	2017	58.0000	0.0000
	2018	57.0000	0.0000
	2019	61.0000	0.0000
	2020	72.0000	0.0000
	2021	80.0000	0.0000
	2022	89.0000	0.0000
	2023	98.0000	93.8000
	2024	97.0000	93.1000
	2025	0.0000	92.0000

Chart: ConvLSTM Actual vs Predicted (Light Tolerant)

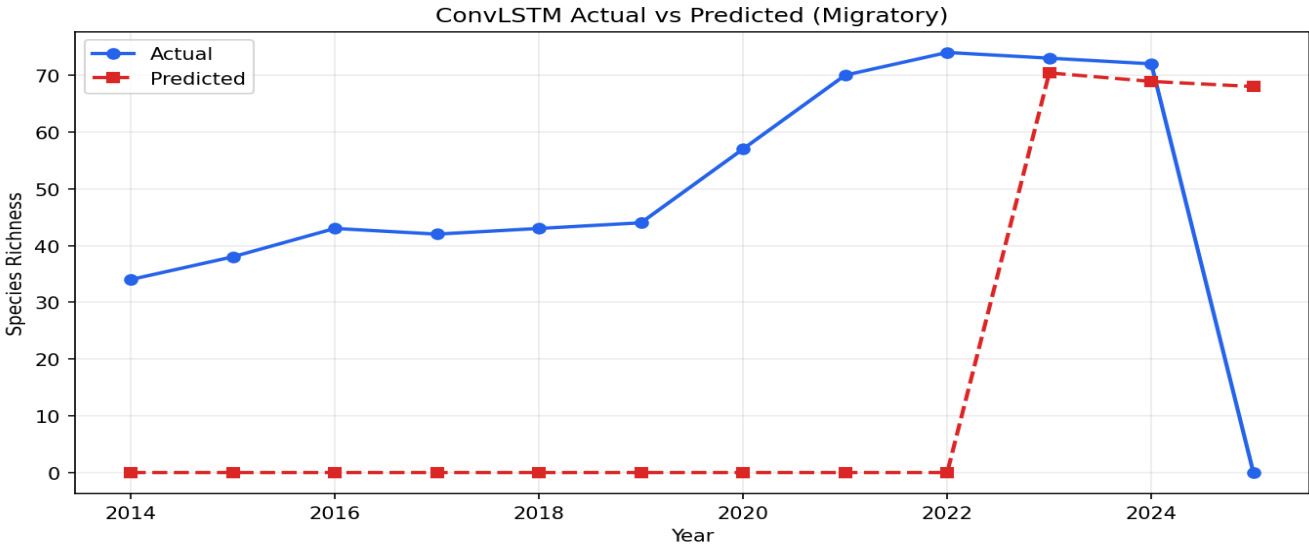


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Numerical Values: ConvLSTM Actual vs Predicted (Light Tolerant)

Year	Actual	Predicted
2014	67.0000	0.0000
2015	74.0000	0.0000
2016	82.0000	0.0000
2017	83.0000	0.0000
2018	83.0000	0.0000
2019	89.0000	0.0000
2020	92.0000	0.0000
2021	92.0000	0.0000
2022	101.0000	0.0000
2023	106.0000	102.1000
2024	102.0000	98.4000
2025	0.0000	98.2000

Chart: ConvLSTM Actual vs Predicted (Migratory)

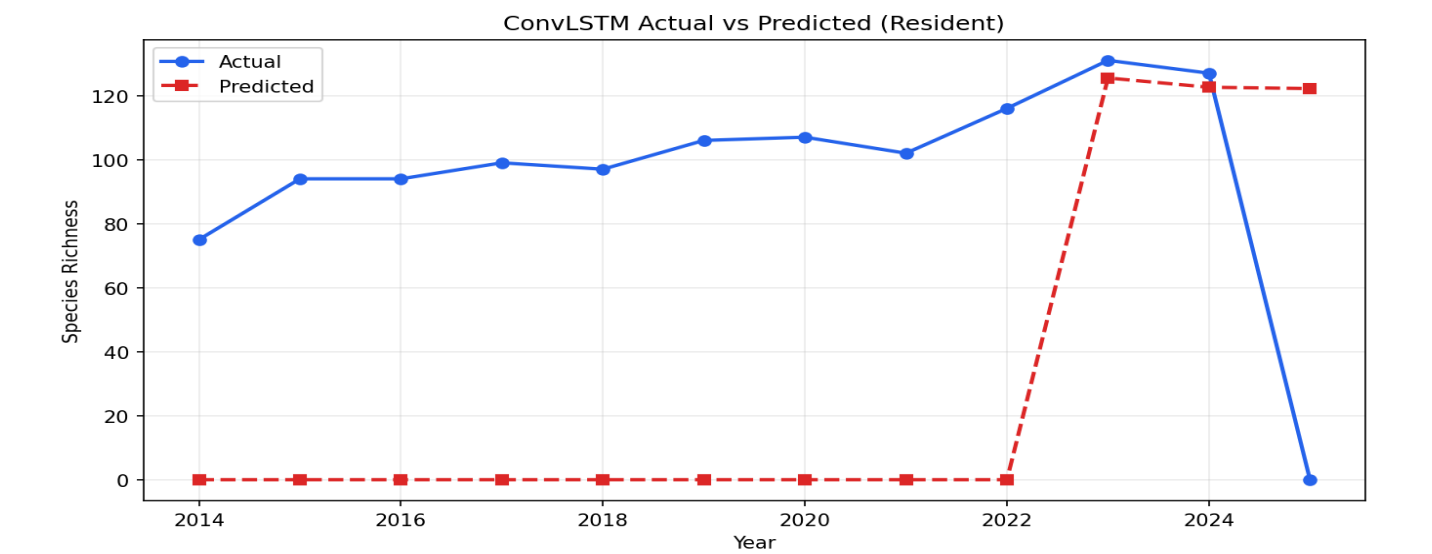


Numerical Values: ConvLSTM Actual vs Predicted (Migratory)

Year	Actual	Predicted
2014	34.0000	0.0000
2015	38.0000	0.0000
2016	43.0000	0.0000
2017	42.0000	0.0000
2018	43.0000	0.0000
2019	44.0000	0.0000
2020	57.0000	0.0000
2021	70.0000	0.0000
2022	74.0000	0.0000
2023	73.0000	70.4000
2024	72.0000	68.9000
2025	0.0000	68.0000

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Chart: ConvLSTM Actual vs Predicted (Resident)



Numerical Values: ConvLSTM Actual vs Predicted (Resident)

Year	Actual	Predicted
2014	75.0000	0.0000
2015	94.0000	0.0000
2016	94.0000	0.0000
2017	99.0000	0.0000
2018	97.0000	0.0000
2019	106.0000	0.0000
2020	107.0000	0.0000
2021	102.0000	0.0000
2022	116.0000	0.0000
2023	131.0000	125.5000
2024	127.0000	122.6000
2025	0.0000	122.2000

Systems Recommendations: KBA/PA Audit (Core)

Site	Type	Species	Sens.Cnt	Sens.%	Light	Eff.	Status
Manila Bay Beach Resort	PA	12	1	8.3	63.4	23.6	Critical
Luneta National Park	PA	29	6	20.7	78.9	26.4	Critical
Ninoy Aquino Parks and Wildlife Center	PA	41	10	24.4	48.2	41.3	At Risk
Las Piñas-Parañaque Wetland Park	PA	64	21	32.8	17.5	60.5	Moderate
Manila Bay	KBA	92	35	38.0	11.6	63.9	Moderate

Systems Recommendations: KBA/PA Audit (Environment)

Site	Mean NDVI	Max LST	Precip	Cells	Snapshot
Manila Bay Beach Resort	0.1952	36.69	209.70	45	2025-12
Luneta National Park	0.2073	37.42	27.01	8	2025-12
Ninoy Aquino Parks and Wildlife Center	0.3363	37.42	39.98	9	2025-12
Las Piñas-Parañaque Wetland Park	0.2970	36.23	198.71	30	2025-12
Manila Bay	0.2370	37.42	4151.56	874	2025-12